

Platform Engineering Hands-On Lab

Progress & Roadmap — August–September 2026

Lab Start	August 29, 2026	Main Build Target	September 12, 2026
Current Phase	Swarm Orchestration	Final Proof / Docs Target	September 16, 2026

Completed Work · Aug 29 – Sep 1

Date	Phase	What We Covered	Status
Aug 29	Lab Architecture	Defined Platform Engineering lab goals; separated Terraform, Ansible, workloads, CI/CD, observability and documentation responsibilities	✓ Done
Aug 29	Repository	Created platform-lab Git repository and directory structure; .gitignore; initial commit	✓ Done
Aug 29	Terraform Foundation	Terraform CLI verification; versions.tf; DigitalOcean provider; provider version constraints; terraform init; .terraform.lock.hcl	✓ Done
Aug 30	Terraform Authentication	DigitalOcean least-privilege API token; environment-variable authentication; provider/data-source concepts	✓ Done
Aug 30	Terraform Data Sources	digitalocean_regions; terraform plan; terraform console -plan; Terraform expressions/filtering; confirmed blr1	✓ Done
Aug 31	Terraform Networking	Created DigitalOcean VPC 10.10.10.0/24; variables; plan/apply; Terraform state fundamentals	✓ Done
Aug 31	Terraform Compute	Existing SSH-key lookup (m4-air); created Ubuntu 24.04 Droplet; dependency graph; tags; provider-computed attributes	✓ Done
Aug 31	Terraform IAM Lesson	Hit missing tag:create permission; refined PAT scopes instead of granting blanket access	✓ Done
Aug 31	Terraform Outputs	Public IP, private IP, Droplet ID and VPC ID exposed as Terraform outputs	✓ Done
Aug 31	Cloud Firewall	Terraform-managed DigitalOcean firewall: SSH restricted to personal /32, HTTP/HTTPS public, outbound allowed	✓ Done
Aug 31	Infrastructure Validation	Verified public/private NICs and VPC IP; SSH access through Terraform-created infrastructure	✓ Done
Aug 31	Ansible Foundation	Inventory, host groups, ansible ping, Python interpreter, facts, ad-hoc commands	✓ Done
Aug 31	Configuration Management	First baseline.yml; /opt/platform; demonstrated Ansible idempotency (changed=1 → changed=0)	✓ Done
Aug 31	Linux Access Model	Created platform admin user; SSH public key; sudo; moved Ansible away from direct root login	✓ Done
Aug 31	SSH Hardening	Disabled root SSH, passwords and keyboard-interactive auth; retained public-key auth; validated sshd before reload	✓ Done
Aug 31	Ansible Roles	Refactored monolithic playbook into common, ssh_hardening, and later docker roles; added ansible.cfg	✓ Done
Aug 31	Docker Automation	Added official Docker APT repository; Docker Engine, CLI, containerd, Buildx and Compose plugin installed via Ansible	✓ Done

Date	Phase	What We Covered	Status
Sep 1	Docker Runtime	Examined dockerd, containerd, Docker socket, Docker group and root-equivalent implications of docker.sock	✓ Done
Sep 1	Container Lifecycle	hello-world, Nginx container, image vs container, stopped/exited/running state, host-port publication	✓ Done
Sep 1	Docker Networking	Default bridge network; container IPs; demonstrated failure of name resolution on default bridge	✓ Done
Sep 1	User-Defined Bridge	Created lab-net; Docker DNS; container-name resolution; busybox → lab-nginx worked by name	✓ Done
Sep 1	Compose Networking Problem	Separate backend and Redis Compose stacks; separate default networks; reproduced cross-stack communication failure	✓ Done
Sep 1	Shared Compose Network	External shared bridge network; backend/Redis communication; Redis test returned PONG	✓ Done
Sep 1	Container Immutability Lesson	Manual apk add redis disappeared after container recreation; distinguished image-defined state from container drift	✓ Done
Sep 1	Swarm Foundation	Ansible Swarm manager role; private VPC advertise/listen/data path address; initialized single-node Swarm	✓ Done
Sep 1	Swarm Verification	platform-node-01 became Ready / Active / Leader; understood manager/worker capability	✓ Done
Sep 1	Swarm Services	Created replicated lab-web service; service vs task vs container; desired-state concept	✓ Done
Sep 1	Reconciliation	Began failure/recovery exercise: kill service task and let Swarm restore desired replica count	● Current

Active Build · Sep 2 – 12

Date	Phase	Target
Sep 2	Swarm Fundamentals	Finish reconciliation; scale replicas 3 → 6 → 2; service lifecycle
Sep 2	Swarm Networking	Overlay networks, service discovery and internal service load balancing
Sep 3	Terraform Multi-Node	Provision platform-node-02 through Terraform; VPC/firewall integration
Sep 4	Ansible Multi-Node	Configure second node automatically; join it to Swarm as worker
Sep 5	Scheduling	Multi-node replica placement, constraints, drain/activate operations
Sep 6	Failure Engineering	Kill containers, stop Docker, lose worker node, observe recovery behavior
Sep 7	Stateful Workloads	Redis persistence, volumes, placement, node-local storage limitations
Sep 8	Production Deployments	Health checks, rolling updates, failed rollout and rollback
Sep 9	Registry	Versioned images, image registry and immutable deployment artifacts
Sep 10	CI/CD	GitHub Actions → test → build → registry → Swarm deployment
Sep 11	Ingress & Security	Reverse proxy, routing mesh, TLS, Swarm firewall rules, secrets
Sep 12	Observability	Metrics, logs, health visibility and integrated platform validation

Proof & Portfolio Phase · Sep 14 – 16

Date	Phase	Target
Sep 14	Destruction Test	terraform destroy; remove the complete lab
Sep 15	Rebuild Test	Recreate from Git using Terraform → Ansible → Swarm → workloads
Sep 16	Portfolio & Runbooks	Architecture diagrams, migration runbook, failure report, README and interview/Upwork material

The Progression

The tools are secondary — this is a single narrative of increasing platform maturity, from static infrastructure to a self-healing, observable system:

Aug 29	■ Architecture & Git
	■ Terraform fundamentals
	■ Cloud networking & compute
	■ Ansible configuration management
	■ Linux security baseline
	■ Docker runtime
	■ Docker networking
	■ Compose limitations
Sep 1	■ Swarm orchestration ← WE ARE HERE
	■ Multi-node platform
	■ Failure recovery
	■ State & storage
	■ Rolling deployment
	■ Registry
	■ CI/CD
	■ Ingress
	■ Observability
Sep 16	■ Destroy → rebuild → document

Platform Engineering Hands-On Lab — Progress & Roadmap, Aug–Sep 2026. Sections: Completed Work (Aug 29–Sep 1), Active Build (Sep 2–12), Proof & Portfolio Phase (Sep 14–16).